



Faculty of Engineering and Technology
Electrical and Computer Engineering Department
DATA BASE SYSTEMS
COMP333
PHASE 2

I-Tech Electronics Store Management System

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Project Scope and Description

1. System Overview

This project is about developing a database application for I-Tech, which is a local electronics store. The store sells different types of products such as smartphones, laptops, and accessories. The main goal of the system is to help organize and manage the daily operations of the store in a clear and efficient way.

2. Core System Functions

The system mainly focuses on organizing product information such as product details, categories, and brands. In addition, it helps track suppliers and customers, while also supporting both sales and purchase transactions.

3. Storage and Organization

The store contains different storage areas, such as a showroom at the front and a storage room, along with other spaces. These areas are divided into multiple storage sections. Each section represents a physical place where products are stored and organized to make access easier.

4. Warranty Management

The system also supports product warranties. Some products, especially devices like smartphones and laptops, may have a warranty, while others, such as some accessories, may not. Warranty information is stored separately and linked only to the products that have a warranty.

5. Technology and Objectives

The system will be developed using Java for the application and MySQL for the database. The main objective is to reduce errors, organize data properly, and make the store's daily work easier and more efficient..

System Requirements

1. Functional Requirements

The system is expected to support a set of main features like:

- Add, update, delete, and view products
- Manage product categories, suppliers, and customers
- Record purchase and sales transactions
- Track inventory quantities
- Generate reports related to stock, sales, and purchases

2. Non-Functional Requirements

The system should meet the following requirements: easy to use, accurate and reliable, connect properly to MySQL, also should support a graphical user interface using JavaFX, and handle daily store operations efficiently

Information about the Client

1. Client Overview

The client is a local electronics store called I-Tech, located in one of the towns in Ramallah. The store sells a variety of electronic products such as smartphones, laptops, chargers, headphones, keyboards, mice, and other accessories.

2. Contact Information

The store can be contacted through the following:

- Phone: (+970-567200596)
- Email: itech@gmail.com

3. Client Needs

The client needs a system that helps organize product data, manage supplier and customer information, track stock levels, and record both sales and purchase transactions. The system should also provide useful reports that help in managing the business more effectively.

NOTE: The entities, attributes, and relationships included in this phase represent an initial design of the system. These elements are not final and may be modified or improved later based on the project requirements and implementation needs.

Entities:

The system is designed based on several entities that represent the main components of the electronics store. These entities were identified according to the system requirements and the operations performed within the store.

Product Entity:

Attribute	Description
Product_ID (PK)	Unique identifier for each product
Category_ID (FK)	Refers to the category of the product
Brand_ID (FK)	Refers to the brand of the product
Product_Name	Name of the product
Model	Product model
Price	Selling price

This entity represents all products available in the store.

Category Entity:

Attribute	Description
Category_ID (PK)	Unique identifier
Category_Name	Name of the category

Used to classify products.

Supplier Entity:

Attribute	Description
Supplier_ID (PK)	Unique identifier for the supplier
Supplier_Name	Name of the supplier
Company_Name	Company name that the supplier work with
Phone	Phone for contact
Address	Address of the supplier

Stores supplier info.

Customer Entity:

Attribute	Description
Customer_ID (PK)	Unique identifier for customers
Customer_Name	Name of the customer
Phone	Phone for contact

Stores customer info.

Storage_Section Entity:

Attribute	Description
Section_ID (PK)	Unique identifier for the section
Section_Name	Name of storage section
Section_Type	Type (like showroom or Backroom)

Represents the different storage areas in the store, such as the showroom, backroom, and other sections where products are stored and organized.

Purchase Entity:

Attribute	Description
Purchase_ID (PK)	Unique identifier for purchase transaction
Supplier_ID (FK)	Foreign key from supplier entity
Purchase_Date	Date of purchase transaction
Delivery_Date	The delivery date of it
Total_Amount	Total cost of the purchase transaction

Represents purchase transactions and the suppliers they are made from.

Sale Entity:

Attribute	Description
Sale_ID (PK)	Unique identifier for sale transaction
Customer_ID (FK)	Foreign key from Customer entity
Sale_Date	Date
Total_Amount	Total

Represents sale transaction and the customer which are made for.

Employee Entity:

Attribute	Description
Employee_ID (PK)	Unique identifier for the Employee
Employee_Name	Name of Employee
Position	Working position of Employee
Salary	Employee salary
Phone	Phone number of Employee

Stores employees working in the store.

Brand Entity:

Attribute	Description
Brand_ID (PK)	Unique identifier for Brand
Brand_Name	Name of the Brand
Website	The website of the Brand

Represents product brands, such as Apple, Samsung, and other manufacturers.

Sale_Details Entity:

Attribute	Description
Sale_ID (FK)	Foreign key from Sale entity
Product_ID (FK)	Foreign key from Product entity
Quantity	Quantity of product in the sale transaction
Unit_Price	The price of unit

Represents the items included in each sale, including the product, quantity, and unit price.

Purchase_Details Entity:

Attribute	Description
Purchase_ID (FK)	Foreign key from Purchase entity
Product_ID (FK)	Foreign key from Product entity
Quantity	Quantity of product in the Purchase transaction
Unit_Price	The price of unit

Represents the products included in each purchase, along with their quantities and unit prices. Each record corresponds to a specific product within a specific purchase.

Warranty (Weak Entity):

Attribute	Description
Product_ID	The Primary key from owner entity (Product)
Warranty_Number	Partial key for the warranty
Warranty_Type	Type of warranty
Warranty_Period	Period time of warranty
Start_Date	Start date of warranty

Represents warranty information for products that include a warranty, such as its type and duration. Each warranty is associated with a specific product and cannot exist independently (because it is weak entity).

Storage_Details Entity:

Attribute	Description
Product_ID (FK)	Foreign key from Product entity
Section_ID (FK)	Foreign key from Storage_section entity
Quantity	Quantity of product stored

Represents the current stock quantity of the product.

Supplier_Product Entity:

Attribute	Description
Supplier_ID (FK)	Foreign key from Supplier entity
Product_ID (FK)	Foreign key from Product entity
Supply_Price	Price of product

Represents the products supplied by each supplier along with their supply prices.

Payment Entity:

Attribute	Description
Payment_ID (PK)	Unique identifier of payment transaction
Sale_ID (FK)	Foreign key from Sale entity
Payment_Date	Date of payment
Amount	Payment amount
Payment_Method	Method of payment (like cash or installment)
Payment_Status	Payment status (paid or pending)

Represents payments made by customers for sales transactions, including details such as payment method and status. Each payment is associated with a specific sale.

Return_Record Entity:

Attribute	Description
Return_ID (PK)	Unique identifier for each return record
Sale_ID (FK)	Part of a composite foreign key referencing Sale_Details
Product_ID (FK)	Part of a composite foreign key referencing Sale_Details
Return_Date	Date of return
Quantity	Returned quantity
Return_Reason	Reason for return
Return_Status	Status of the return (like: approved, pending)

Represents products that are returned by customers after a sale, including details such as quantity, reason, and status. Each return is associated with a specific product within a specific sale.

Stock_Movement_History Entity:

Attribute	Description
Movement_ID (PK)	Unique identifier for each stock movement
Section_ID (FK)	Foreign key from storage_section entity
Product_ID (FK)	Foreign key from Product entity
Movement_Type	Type of movement (like: purchase, sale, return, transfer)
Movement_Date	Date of movement
Quantity	Quantity of product moved

Stores records of all stock movements, including the type, date, and quantity. Each entry represents a specific movement of a product.

Relationships:

Relationships Overview

Entities	Relationship	Description	Type
Category & Product	belongs to	Each category can include many products and each product belongs to only one category	1:M
Supplier & Supplier_Product	supplies	A supplier can be associated with many supplier-product records and each supplier-product record belongs to one supplier	1:M
Product & Supplier_Product	is supplied by	A product can be associated with many supplier-product records and each supplier-product record belongs to one product	1:M
Customer & Sale	makes	A customer can make many sales transactions and each sale is made by one customer	1:M
Sale & Sale_Details	includes	A sale can include multiple detail records, and each record belongs to one sale.	1:M
Product & Sale_Details	appears in	A product can appear in multiple sale detail records and each sale detail record refers to one product.	1:M
Purchase & Purchase_Details	includes	Each purchase can include multiple purchase detail records and each purchase detail record belongs to one purchase.	1:M
Product & Purchase_Details	appears in	A product can appear in multiple purchase detail records and each purchase detail record refers to only one product.	1:M
Supplier & Purchase	provides	A supplier can be associated with many purchase transactions and each purchase is linked to one supplier.	1:M
Employee & Employee	supervises	An employee can supervise multiple employees, while an employee may or may not have a supervisor (Recursive Relationship)	1:M
Brand & Product	produces	Each brand can produce many products and each product belongs to one brand.	1:M
Product & Warranty	has	Each product can have one warranty and each warranty belongs to one product.	1:1
Storage_Section & Storage_Details	contains	Each storage section can contain multiple storage detail records and each storage detail record belongs to one storage section.	1:M
Product & Storage_Details	is stored in	A product can appear in multiple storage detail records and each storage detail record refers to one product.	1:M

Relationships Overview (Continued)

Entities	Relationship	Description	Type
Sale & Payment	has	Each sale can have one or more payments and each payment belongs to only one sale.	1:M
Sale_Details & Return_Record	has	Each sale detail record can have zero or more return records and each return record belongs to one sale detail record.	1:M
Product & Stock_Movement_History	has	A product can have multiple stock movement records and each stock movement record belongs to one product.	1:M
Storage_Section & Stock_Movement_History	records	A storage section can have multiple stock movement records and each stock movement record belongs to one storage section.	1:M

Database Design Analysis:

1.Total Participation

1) **Product & Warranty**

Total participation from Warranty side : All warranties must be associated with a product

2) **Customer & Sale**

Total participation from Sale side: Every sale must be made by a customer

3) **Supplier & Purchase**

Total participation from Purchase side: Every purchase must be linked to a supplier

4) **Sale & Payment**

Total participation from Payment side: Every payment must belong to a sale

5) **Sale_Details & Return_Record**

Total participation from Return_Record side: Every return must be associated with a sale detail record

6) **Product & Stock_Movement_History**

Total participation from Stock_Movement_History side: Every stock movement must be related to a product

7) **Storage_Section & Stock_Movement_History**

Total participation from Stock_Movement_History side: Every stock movement must be associated with a storage section

8) **Sale & Sale_Details**

Total participation from Sale_Details side: Every sale detail record must belong to a sale

9) **Product & Sale_Details**

Total participation from Sale_Details side: Every sale detail record must refer to a product

10) **Purchase & Purchase_Details**

Total participation from Purchase_Details side: Every purchase detail record must belong to a purchase

11) **Product & Purchase_Details**

Total participation from Purchase_Details side: Every purchase detail record must refer to a product

12) **Supplier & Product_Supplier**

Total participation from Product_Supplier side: Every supplier-product record must be associated with a supplier

13) **Product & Product_Supplier**

Total participation from Product_Supplier side: Every supplier-product record must be associated with a product

14) **Storage_Section & Storage_Details**

Total participation from Storage_Details side: Every storage detail record must belong to a storage section

15) **Product & Storage_Details**

Total participation from Storage_Details side: Every storage detail record must refer to a product

16) **Category & Product**

Total participation from Product side: Every product must belong to a category

Note: All other relationships in the diagram have partial participation.

2. Recursive Relationship

The system includes a recursive relationship in: **Employee & Employee**. Since an employee can supervise multiple employees. So each employee may or may not have a supervisor. The type of the relation is One-to-Many (recursive) with partial participation.

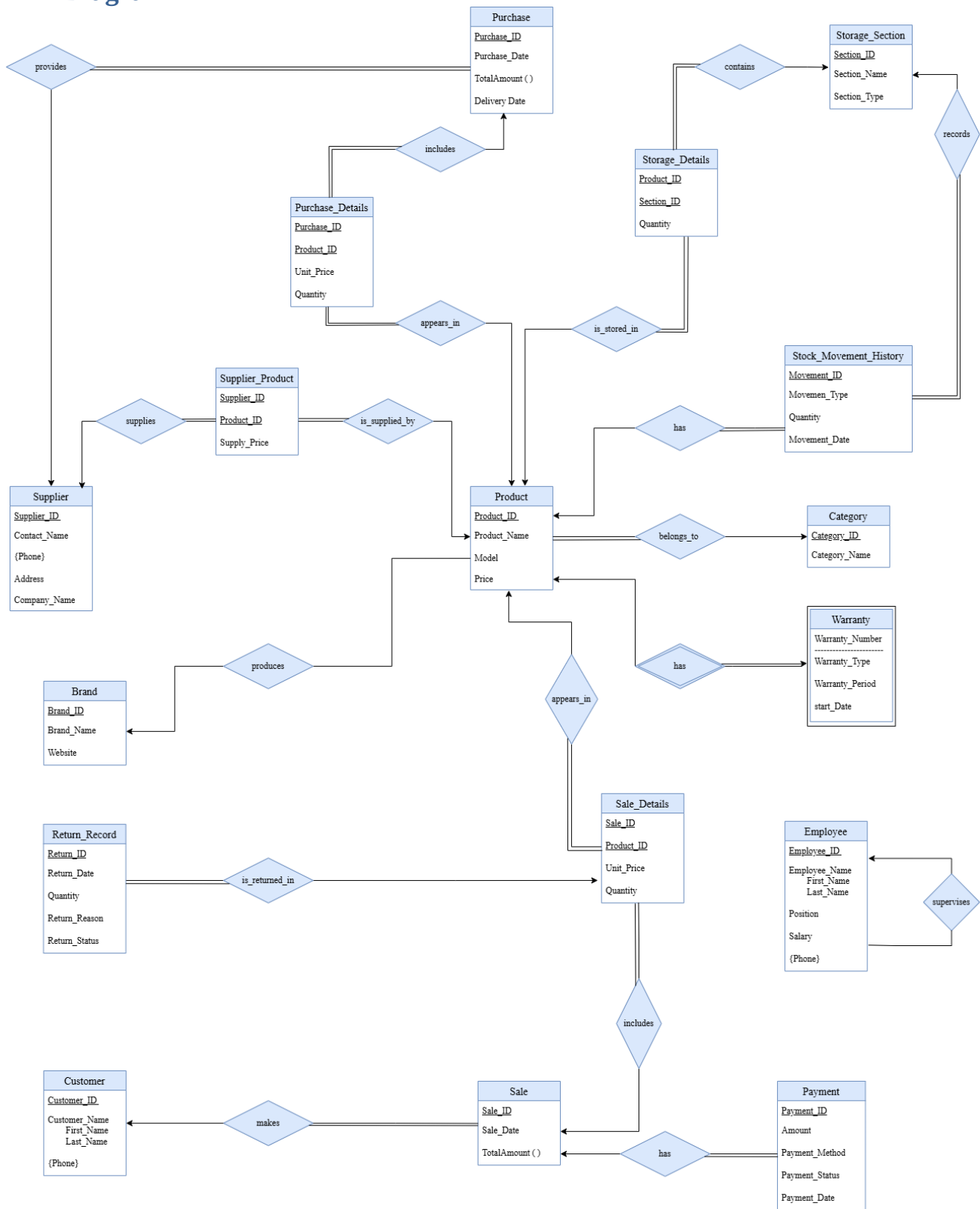
3.Non-Binary Relationships

The current system does not require ternary relationships because all requirements can be represented using binary relationships.

4.Class Hierarchy

Although class hierarchies were considered, the system uses a Category entity to classify products, which provides a simpler and more suitable design for this project.

ER Diagram



ER Diagram Explanation

1. General Explanation

The ER diagram shows the general structure of the database for the electronics store. It explains how the main entities like Product, Sale, Purchase, Customer, and Supplier are connected together using different relationships.

Also, it helps to understand how the data moves inside the system, like in sales, purchases, managing stock, and returns. So basically it gives a full picture of how everything is related.

2. Important Design Decisions

While working on the ER diagram, we made some important decisions to make the design clearer and more organized.

First, the many-to-many relationships (like between Sale and Product, Purchase and Product, Supplier and Product, and Storage_Section and Product) were converted into separate tables. These tables are Sale_Details, Purchase_Details, Product_Supplier, and Storage_Details. This was needed because we also had extra attributes like quantity and price.

Another thing we did was linking Return_Record with Sale_Details instead of linking it directly with Sale and Product. This makes more sense because each return actually belongs to a specific product inside a specific sale, not just the sale in general.

Also, Warranty was designed as a weak entity, since it depends on Product and cannot exist alone. In addition, some attributes were modeled differently based on their nature. For example, phone numbers were considered as multi-valued attributes, since a supplier or customer can have more than one phone number. Also, some values such as total amounts can be derived from other attributes, so they were treated as derived attributes..

Finally, we used total participation in some cases where it is required that an entity must be related to another one. For example, every Sale must have a Customer, and every Payment must be linked to a Sale.

3. Relationships Explanation

The relationships in the diagram show how the entities interact with each other.

For example, a Customer can make more than one Sale, and each Sale can include multiple products using Sale_Details. Products are also connected to Category and Brand.

Also, stock movement is handled using Stock_Movement_History, which is connected to both Product and Storage_Section to track changes in inventory.

Sample Queries

1. Retrieve product names, categories, brands, and prices for all products, sorted by category and product name.
2. Retrieve product names, brands, prices, and stock quantities of all products supplied by a specific supplier, sorted by product name.
3. Retrieve total stock quantity for each category, sorted by total quantity in descending order.
4. Retrieve product names, categories, brands, and stock quantities of all products with low stock quantity, sorted by stock quantity.
5. Retrieve sale IDs, customer names, sale dates, and total amounts for all sales made on a specific date, sorted by total amount.
6. Retrieve product names, categories, brands, and stock quantities of all products stored in a specific storage section, sorted by product name.
7. Retrieve product names and total sold quantities of the most sold products, sorted by sold quantity in descending order.
8. Retrieve customer names and the total value of their sales transactions.
9. Retrieve product names, categories, brands, and prices of products that were never sold, sorted by product name.
10. Retrieve the average price of products in each category, sorted by average price in descending order.
11. Retrieve total sales amount per day.
12. Retrieve purchase ID, purchase date, supplier name, product names, quantities, and total amount for a specific purchase transaction.
13. Retrieve product names, categories, models, prices, and stock quantities of all products of a specific brand, sorted by category and price.
14. Retrieve the top 5 brands with the highest number of products.
15. Retrieve sale IDs, sale dates, and total amounts for all sales made by a specific customer, sorted by sale date.
16. Retrieve supplier names and phone numbers of all suppliers, sorted by supplier names.
17. Retrieve product names, categories, brands, and prices of all products with price less than a specific value, sorted by price.
18. Retrieve employee names, positions, salaries, and phone numbers of all employees supervised by a specific manager, sorted by employee name.
19. Retrieve the average salary of all employees.
20. Retrieve employee names, positions, and salaries of employees who earn above the average salary, sorted by salary in descending order.
21. Retrieve the most expensive product in each category, including product name, category, and price.